Input system:
Dependent variables:$p_1, g_2, p_3, g_3, g_1, p_2$
Independent variables:t
Equations:

$$\frac{dg_1}{dt} = p_1$$

$$\frac{dg_2}{dt} = p_2$$

$$\frac{dg_3}{dt} = p_3$$

$$\frac{dp_1}{dt} = (-1)\, k^2\, (m_0 + m_1)\, g_1 \mathrm{isqrt}\left(g_1^2 + g_2^2 + g_3^2\right)^3$$

$$\frac{dp_2}{dt} = (-1)\, k^2\, (m_0 + m_1)\, g_2 \mathrm{isqrt}\left(g_1^2 + g_2^2 + g_3^2\right)^3$$

$$\frac{dp_3}{dt} = (-1)\, k^2\, (m_0 + m_1)\, g_3 \mathrm{isqrt}\left(g_1^2 + g_2^2 + g_3^2\right)^3$$

==========

Result:
Dependent variables:$s_1, p_3, g_1, g_3, p_1, g_2, p_2$
Independent variables:t
Additional variables:

$$s_1 = \mathrm{isqrt}\left(g_1^2 + g_2^2 + g_3^2\right)$$

Equations:

$$\frac{dg_1}{dt} = p_1$$

$$\frac{dg_2}{dt} = p_2$$

$$\frac{dg_3}{dt} = p_3$$

$$\frac{dp_1}{dt} = (-1)\, s_1^3 g_1 k^2 m_0 + (-1)\, s_1^3 g_1 k^2 m_1$$

$$\frac{dp_2}{dt} = (-1)\, s_1^3 g_2 k^2 m_0 + (-1)\, s_1^3 g_2 k^2 m_1$$

$$\frac{dp_3}{dt} = (-1)\, s_1^3 g_3 k^2 m_0 + (-1)\, s_1^3 g_3 k^2 m_1$$

$$\frac{ds_1}{dt} = (-1)\, s_1^3 g_1 p_1 + (-1)\, s_1^3 g_2 p_2 + (-1)\, s_1^3 g_3 p_3$$